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MILITARY SPECIFICATION

TEST STAND, HYDRAULIC SYSTEM COMPONENTS, TYPE MK-3A (STAND, TEST, AIRCRAFT HYDRAULIC SYSTEM, ELECTRIC MOTOR DRIVEN, TYPE MK-3A)

1. SCOPE

1.1 This specification covers one type of electric motor driven, aircraft hydraulic system test stand, designated Type MK-3A.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

CC-M-636	Motor, Alternating-Current, (Fractional Horsepower)
CC-M-641	Motor, Alternating Current, (Integral Horsepower, 200 HP And Smaller)
GG-G-76	Gages, Pressure And Vacuum, Dial Indicating, (For Air, Steam, Oil, Water, Ammonia And Chloro-Fluoro Hydrocarbon Gases)
PPP-T-60	Tape: Pressure-Sensitive Adhesive; Waterproof, For Packaging

Military

MIL-C-104	Crates, Wood; Lumber And Plywood Sheathed, Nailed And Bolted
MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-5440	Hydraulic Systems, Aircraft, Types I And II, Design, Installation, And Data Requirements For
MIL-R-5520	Reservoirs: Aircraft, Hydraulic, Non-Separated Type
MIL-H-5606	Hydraulic Fluid, Petroleum Base: Aircraft, Missile, And Ordnance
MIL-C-5756	Cable And Wire, Power, Electric, Portable

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MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification For
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-F-27656	Filter Unit, Fluid, Pressure MXU-408/M, Absolute 5 Micron, Hydraulic
MIL-S-38159	Connectors, Electric, Ramp Power, 416/240 Volts
MIL-C-45662	Calibration System Requirements

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order of Precedence For The Selection Of
MIL-STD-470	Maintainability Program Requirements (For Systems And Equipments)
MIL-STD-808	Finishes, Protective, And Codes, For Finishing Schemes For Ground And Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements And Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS25331	Light Assembly, Press To Test Indicator
MS27265	Connector, Plug, Electrical-Ramp Power, 416/240 Volts, Male
MS27766	Shroud For Aerospace Ground Equipment
MS28720	Filter, Aircraft Hydraulic Line Type, 3,000 PSI
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS28759	Hose Assembly, Rubber, Hydraulic, (3,000 PSI), Flared Tube
MS28777	Washer, Aircraft Hydraulic Packing Back-Up
MS28778	Packing, Preformed, Straight Thread Tube Fitting Boss
MS33649	Bosses, Fluid Connection-Internal Straight Thread
MS33656	Fitting End, Standard Dimensions For Flared Tube Connection And Gasket Seal
MS33657	Fitting End, Standard Dimensions For Bulkhead Flared Tube Connection
MS51335	Pintle Assembly, Towing, 18,000 LBS. Capacity, Manual Release

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Air Force-Navy Aeronautical

AN832	Union-Flared Tube, 3/8 Bulkhead And Universal
AN910	Coupling - Pipe Thread
AN924	Nut, Plain Hexagon, Tube, Bulkhead And Universal Fitting
AN929	Cap Assembly, Pressure Seal Flared Tube Fitting
AN6235	Filter Element - Hydraulic Replaceable Micronic Line Type
AN6236	Filter Element - Hydraulic Replaceable Micronic Reservoir Type
AN6247	Valve - Controllable Hydraulic Check (Rotating Action)
AN6249	Valve, 3000 PSI Hydraulic Check
AN6270	Hose Assembly - Detachable Swivel Fitting, Low Pressure
AN6289	Nut - Flared Tube Universal Fitting
AND10260	Flange - Type X Accessory Mounting
AND10261	Flange - Type XI Accessory Mounting
AND10266	Flange Type XVI Accessory Mounting
AND20006	Drive Type XVI Engine Accessory

DRAWINGS

Air Force

48B7796	Ring Assembly - Tie Down 10000 Lb
49B6564	Ring - Eyebolt 25000 Lb Cargo Tiedown

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION

Pamphlet 70 National Electrical Code

(Copies of the NFPA pamphlet may be obtained upon application to the National Fire Protection Association, 60 Batterymarch Street, Boston MA 02110.)

UNDERWRITERS' LABORATORIES, INC

Standard for Safety

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island NY 11746.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

MG1 Standards For Motors And Generators

(Copies of the NEMA standard may be obtained upon application to the National Electrical Manufacturers Association, Engineering Department, 155 East 44th Street, New York NY 10017.)

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AMERICAN STANDARDS ASSOCIATION

ASA C50 Rotating Electrical Machinery

(Copies of the ASA standard may be obtained upon application to the American Standards Association, 10 East 40th Street, New York NY 10016.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Pressure Vessels (Section VIII)

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East Forty-Seventh Street, New York NY 10017.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The test stand shall consist of the following major components:

	Description	See Requirement
a.	Hydraulic system	3.7
b.	Electrical sytem	3.8
c.	Pneumatic system	3.9
d.	Fan or blower	3.10
e.	Trailer and housing	3.11

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143

3.4 Materials.

3.4.1 Protective treatment. When materials are used in the construction of the test stand that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The test stand shall be a dual-system, self-contained, mobile testing unit designed to provide not less than 54 gallons per minute flow at 2,750 pounds per square inch gage. The components shall be arranged as shown on Figure 1. The stand shall also provide not less than 30 gallons per minute at 4,600 pounds per square inch gage.

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* 3.5.1 Reliability. The test stand shall be designed and constructed as specified herein and to complete the tests of 4.4.3 successfully with no failures preventing satisfactory performance. These tests shall demonstrate a mean time between failure (MTBF) of at least 152 hours when calculated in accordance with 4.4.2.1 The mean life or MTBF is the arithmetical mean (average) of the operating time between failures. A factor of 2/3 multiplies the test MTBF as an added assurance.

* 3.5.2 Maintainability. The design of the test stand shall reflect the intent of MIL-STD-470 which will be demonstrated during disassembly inspection per 4.6.22. Specific attention shall be given to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing.

c. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to procuring activity approval.

d. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location.

e. Minimum number of tools required for maintenance by design practices such as reducing the number of different bolt head sizes to the practicable minimum.

f. The maximum time to remove any component, including the 60 horsepower motor, shall not exceed 6 hours. Mean time to repair shall not exceed 2 man-hours.

3.5.3 Safety. The test stand shall be constructed to provide for maximum safety to personnel operating and maintaining the unit.

* 3.5.3.1 Static electricity. Provisions shall be made for static electricity discharge from sight indicators, flowmeters and from the test stand itself.

3.5.4 Operating and service controls. All operating and service controls shall be readily accessible to and operable by an operator wearing heavy gloves.

3.6 Performance. The test stand shall be capable of operating under and withstanding the following conditions:

a. Operating temperatures ranging from -40° to +125° Fahrenheit (F).

b. Pressure altitudes from sea level to 6,000 feet.

c. A relative humidity of 95 to 100 percent at an operating temperature of 125°F and a storage temperature of 160°F.

d. Fungus growth as encountered in tropical climates.

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e. Sand and dust particles as encountered in desert areas.

f. Rainfall as encountered in any locale.

g. Exposure to salt-sea atmosphere.

3.6.1 Storage temperatures. The test stand shall be capable of withstanding storage temperatures ranging from -65° to +160°F.

3.6.2 Operating positions. The test stand shall function satisfactorily when its normal operating plane is at an angle of 8-1/2 degrees in any direction from horizontal.

3.6.3 Noise level. The noise level of the test stand shall not exceed the values in the frequency ranges specified in 4.6.14.

* 3.7 Hydraulic system. Unless otherwise specified, the requirements of MIL-H-5440 apply in their entirety with regard to materials, fluids, reservoir venting and the cleaning of parts and systems. With regard to tubing, MIL-H-5440 will be followed with respect to material, tubing bends, tubing support, routing, looping, undue strain on tubing, clamping, spacing and flaring.

3.7.1 High-pressure pumps. Two axial-piston-type, high-pressure pumps shall be provided. They shall have aircraft pump impulse characteristics by using nine pistons and high operational speeds (1,500 to 4,000 revolutions per minute inclusive). The pumps shall be of the variable-stroke, variable-pressure, pressure-compensated type with adjustable maximum volume control. The pumps shall be of sufficient rating to meet all requirements specified herein but shall be not less than 30 gallons per minute at 3,000 pounds per square inch gage at 3,750 revolutions per minute and 15 gallons per minute at 5,000 pounds per square inch gage at 3,750 revolutions per minute.

* 3.7.1.1 Life expectancy. The high pressure pumps shall have a minimum life expectancy of 1,000 hours between overhauls.

3.7.1.2 Each pump shall require not more than 54 horsepower from the motor specified in 3.8.1 when being driven at 3,600 revolutions per minute and pumping hydraulic fluid conforming to MIL-H-5606 at 27 gallons per minute and raising the fluid pressure from 40 to 3,000 pounds per square inch gage at 68°F.

* 3.7.1.3 Each pump shall withstand and operate throughout inlet fluid temperatures ranging from -40° to 250°F and inlet pressure of 40 to 165 pounds per square inch gage.

3.7.1.4 Pressure compensator control. Each pump shall have an adjustable pressure compensator control for regulating the pressure at a constant value with the flow varying from 0 to 27 gallons per minute. The discharge pressure of the pumps shall be 2,700 (±50) pounds per square inch gage when delivering 27 gallons per minute and when the zero flow pressure setting is at 3,000 pounds per square inch gage. The pumps shall meet the performance characteristics shown on Figure 2 and 3. The pumps shall compensate from zero flow to full flow or vice versa in 0.05 second, or less, at any speed above 50 percent of rated speed and shall limit the maximum transient pressure during cutout to not more than 133 percent of nominal rated pressure. The pump pressure shall

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stabilize to a steady-state value within 10 seconds. The adjustment range of the pressure compensator shall be from 750 to 5,000 pounds per square inch. The variable volume control and pressure control shall be mounted on or be easily accessible from the control panel.

3.7.1.5 Relief valve. A fixed relief valve shall be incorporated if required to maintain the pump case pressure at a value recommended by the pump manufacturer.

3.7.1.6 Flanges and spline drive. The pumps shall be equipped with flanges and spline drive conforming to AND10266, type XVI-C. Rotation shall be counterclockwise as viewed from the shaft end.

3.7.1.7 High-pressure pump protection. A pressure differential switch shall be provided in each system. The switches shall actuate when the differential pressure between the inlet port and the case drain line port of the pump drops below 30 pounds per square inch, or as otherwise recommended by the high-pressure pump manufacturer. The switch actuation shall illuminate an MS25331-9 red indicator light on the control panel. The light shall be wired through a holding relay which holds the light on following a low differential pressure. The switches shall actuate only when the differential pressure drop occurs for 0.25 to 0.75 second. In addition to illuminating the indicator light for the faulty system, the switch shall automatically open the circuit to the main drive motor. The red light for the faulty system shall remain illuminated after the motor stops and the light for the other system shall not illuminate during or after the shutdown. Before restarting the faulty system, it shall be necessary to manually reset the device. The reset device shall be on the control panel.

* 3.7.2 Low-pressure (boost) pumps. Two hydraulic boost pumps shall be provided as shown on Figure 1. The pumps shall have a minimum life expectancy of 1,000 hours. Each boost pump shall supply at least 35 gallons per minute continuously to each high-pressure pump with the inlet pressures at 20 inches mercury vacuum and the outlet pressure not less than 40 nor more than 300 pounds per square inch gage, as recommended by the high-pressure pump manufacturer, at all flow and pressure conditions specified herein. The 20 inches mercury vacuum with the above specified flows and pressures shall be required when fluid temperature is below 20°F and 15 inches mercury vacuum shall be required at all fluid temperatures above 20°F. The boost pump shall have sufficient output flow to provide additional fluid flow for filter bleed operations without substantially reducing the high-pressure pump inlet pressure. Check valves shall be located as shown on Figure 1 to permit fluid to bypass the boost pumps. The check valve shall be a 1-1/2 inch, quick acting, poppet type with a maximum of 2 pounds per square inch pressure drop when passing 27 gallons per minute. The boost pump drive shaft seals shall withstand 200 pounds per square inch inlet pressure, 300 pounds per square inch proof pressure.

3.7.2.1 Mounting flange. The boost pumps shall be equipped with flanges and spline drive conforming to AND10261, type XI-C. Rotation shall be clockwise as viewed from the shaft end.

* 3.7.3 Filters. Filters (air, fill-system, low-pressure, high-pressure, and pump-case drain) shall be provided and located in the systems as shown on Figure 1. Each filter or set of filters shall be protected with a differential pressure switch installed as shown on Figure 1 and shall indicate when the pressure drop across the filter reaches a value at which change of the filter

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element is required. The indication shall be the closing of a switch which illuminates an MS25331-9 red light on the control panel. The filters shall be available through access doors or panels to permit easy draining of the case and replacement of the elements.

* 3.7.3.1 High-pressure filters. Two high-pressure filters rated for operation at 30 gallons per minute and 5,000 pounds per square inch shall be incorporated. The filters shall conform to MIL-F-27656.

* 3.7.3.2 Low-pressure filters. Two low-pressure filters shall use 2 or more replaceable elements dimensionally interchangeable with AN6236-3 but rated for 5 micron nominal particle size removal. The low-pressure filters shall be located on a level above any other part of the line between the boost pump and high-pressure pump so that the air will collect within the filter and not be discharged into the high-pressure-pump inlet line. The filters shall be made from corrosion resistant material.

3.7.3.2.1 Filter-bleed valves. Pushbutton-type bleed valves shall be mounted on the control panel and shall serve to bleed entrapped air to the reservoir from the top of the low-pressure filter case or any other location at which air may be apt to collect. The valves shall also serve to bleed or drain the reservoirs of the aircraft, when desired. Force necessary to operate this valve shall not permit leakage from the aircraft hydraulic system to exceed 2 drops in 5 minutes. Valve and tubing size shall be not less than 3/8 inch tube size.

3.7.3.2.1.1 Sight tubes. Sight tubes shall be provided in the bleed lines as shown on Figure 1. They shall be mounted on the control panel adjacent to their respective bleed valves. The sight tubes shall withstand a 200 pounds per square inch working pressure and a 300 pounds per square inch proof pressure. They shall be 3/8 inch size and 3 inch minimum visible length. The sight tubes shall be lighted from behind for ease of visibility of contamination or entrained air.

3.7.3.3 Pump case filters. Two filters conforming to or dimensionally interchangeable with MS28720-12, suitable for 125 pounds per square inch pressure, containing elements conforming to AN6235-4A shall be provided in the pump case drain lines as shown on Figure 1.

* 3.7.4 Thermal pressure relief valves. Preset relief valves dimensionally interchangeable with AN6249 check valves shall be provided in each hydraulic system as shown on Figure 1. The valves shall relieve pressures in excess of 200 pounds per square inch and thus preclude damage to components in the low-pressure hydraulic circuit.

3.7.5 Relief valves. High-pressure and low pressure relief valves shall be provided and located in the system as shown on Figure 1. The valves shall be located as near the high-pressure pump inlet and outlet as possible. The relief valves may be located on the control panel, but they must be installed so that the operator can adjust and read the pressure gage simultaneously. The relief valves shall incorporate a locking device so that vibration cannot change their settings.

3.7.5.1 High-pressure relief valves. The high-pressure relief valves shall be adjustable by hand from 500 to 5,500 pounds per square inch gage.

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3.7.5.2 Low-pressure relief valves. The low-pressure relief valves shall be adjustable to a fixed pressure of not less than 50 nor more than 150 pounds per square inch gage. The low-pressure relief valves shall close to zero flow in 0.025 second and shall have a pressure drop no greater than 25 pounds per square inch when passing 27 gallons per minute. The low-pressure relief valves shall be sensitive to backpressure.

3.7.6 Reservoir. A stainless steel or aluminum, hydraulic-fluid reservoir shall be mounted within the stand enclosure. The reservoir shall have a sufficient capacity to supply at least 25 gallons of fluid to the systems when inclined 8-1/2 degrees from vertical in any direction. The reservoir shall have additional space for a 3-gallon expansion above the normal full level and at least a 2-gallon space below the outlet to the system for collecting moisture and impurities. The reservoir shall be designed for operating pressures of 50 pounds per square inch and structural proof tests of 100 pounds per square inch. The reservoir shall conform to MIL-R-5520 regarding obstruction to flow, structural design, shell, seams, structural strength, baffles, fluid connections, sump, drain, return line inlet, scuppers, filler unit, finish, and filling time. Sufficient clearance shall be provided around the filler opening to permit rapid filling from 1-gallon cans while the test stand is in operation. The reservoir fill cap shall be either the type that cannot be removed while the reservoir is pressurized, or the type that will automatically depressurize while being removed.

3.7.6.1 Access plates. One or more access plates of at least 8 by 6 inches in accordance with standard design practices for American Society of Mechanical Engineers (ASME) pressure vessel code shall be provided to permit visual inspection and manual cleaning of the interior of the reservoir.

3.7.6.2 Drain valve. A drain valve shall be provided for completely draining the reservoir. Necessary clearance shall be provided around the drain outlet to permit rapid draining into 1 gallon containers.

3.7.6.3 Fluid-level indicator. A dial-type, fluid-level indicator shall be provided and shall indicate the quantity of available fluid in the reservoir to an accuracy of within ±2 gallons of actual quantity. The indicator shall be mounted on the instrument panel.

3.7.6.4 Reservoir selector valves. Four-way, two-position selector valves shall be furnished as shown on Figure 1 to either route the return fluid to and from the stand reservoir or to cause the return fluid to bypass the reservoir. The valves shall be capable of operating in pressures varying between 20 inches mercury vacuum and 200 pounds per square inch. The valves shall be mounted on the instrument panel. Internal leakage shall not exceed 5 drops per minute under any operating conditions.

3.7.7 Stand outlet connections. Outlets shall be provided as shown on Figure 1. The connections shall not be located where leakage of fluid from the connection could fall on the trailer tires or within the enclosure of the stand. The connections shall be color coded.

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* 3.7.7.1 Pressure connections. Each pressure outlet and dust cap shall be equal to and dimensionally interchangeable with Aeroquip part No. TA-155-S4-16D and No. 155-S7-16D, respectively. All valves, connections, and tubing used from the high-pressure pump discharge to the end of the 20-foot pressure hose shall be of a size that will not permit the pressure drop to exceed 125 pounds per square inch with 30 gallons per minute flow at 120°F fluid temperature. All connnections and fittings on the high-pressure side shall be not less than 1-inch tubing size. Tubing shall be used on all high-pressure lines within the stand, except as otherwise specified or where impractical because of vibration. The entire high-pressure portion of the systems shall be designed for proof pressures of 7,500 pounds per square inch gage.

* 3.7.8 Return connections. Return connections shall be provided as shown on Figure 1. The connections shall conform to Aeroquip part No. B145-S4-24D or be equal to and dimensionally interchangeable with the Aeroquip part. A dust cap conforming to Aeroquip part No. 145-S7-24D or equal to and dimensionally interchangeable with the Aeroquip part shall be attached to the outlet. The size of all the connections and components used in the test stand hydraulic system shall be such as to permit not more than 10 pounds per square inch drop at 30 gallons per minute flow with fluid temperatures of 120°F from the aircraft outlet through 10 feet of hose to the suction side of the boost pumps. Return lines and components and low pressure lines and components shall be not less than 1-1/2-inch tubing size.

3.7.8.1 External and internal leakage. Total external leakage shall not exceed 2 drops per hour at any flows or pressures specified herein. Internal leakage shall not exceed 5 cubic inches per hour from the closed loop system of either system.

* 3.7.9 Oil cooler. An air-to-oil cooler or coolers shall be provided in the systems as shown in Figure 1. The coolers shall be capable of maintaining the oil temperature of each system below 140°F when 30 gallons per minute are flowed through the coolers with an ambient air temperature of 100°F. The oil cooler or coolers shall be designed for working pressures of 300 pounds per square inch gage without leakage or damage and shall be proof tested at 150 percent of the working pressure. Means shall be provided to make the oil coolers inoperable in low temperatures. Provisions to accomplish this shall be accessible but need not be on the control panel. A certificate will be furnished with each cooler certifying that the cooler has passed a 300 pounds per square inch proof pressure at 165°F.

* 3.7.9.1 An accumulator shall be installed upstream of the oil cooler to protect it from oil surges.

* 3.7.10 Flow control valves. Two valves, 1 inch or larger, shall control or stop the flow into or from each pressure outlet. The valves shall be rated for 27 gallons per minute to 30 gallons per minute flow and 5,000 pounds per square inch pressure. The valves shall accurately meter the flow throughout the volume range of 0 to 30 gallons per minute and shall be so pressure-balanced that a force of 40 inch pounds is the maximum required turning force at 3,000 pounds per square inch. The valves shall be mounted on the control panel and shall be capable of going from the closed position to the open position in two full turns of the handle. The valves shall be capable of controlling the flow within ±1/4 gallon per minute when flow rates are below 5 gallons per minute, and ±1 gallon per minute with flow rates of from 5 to 30 gallons per minute.

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3.7.11 Bypass valves. Two bypass valves, identical to the flow control valves specified in 3.7.10, shall be provided in the systems as shown in Figure 1. The valves shall be mounted on the control panel.

3.7.12 Manifold valve. A 4-way, 2-position, closed-center, shear-seal-type valve rated at 5,000 pounds per square inch working pressure shall be provided in the system as shown on Figure 1. The valve shall shut off the flow to all four ports in one position and shall connect the two return lines together and the two pressure lines together in the other position. The valve shall not be located on the control panel, but shall be easily accessible.

3.7.13 Fill system. A system incorporating an electric-motor-driven pump, filter, pushbutton-type fill valves, pressure gage, and relief valves as shown on Figure 1 shall be provided to add oil to the aircraft or test stand system as desired.

3.7.13.1 Filling pump and motor. The filling pump and motor combination shall pump hydraulic fluid at a minimum rate of 2 gallons per minute at 150 pounds per square inch gage with fluid temperature of 20°F. The pump motor shall be a 220/440-volt, 3-phase, 60-cycle motor of sufficient rating to operate the pump at maximum rated pressure and output continuously without exceeding safe temperature limits for the class of insulation used. Overload relays shall be provided for protecting the pump motor. The pump motor shall be connected as shown on Figure 4. The pump shall be equipped with flange and spline drive conforming to AND10260 type X. The pump shall be driven at 3,600 revolutions per minute.

3.7.13.2 Filling valves. Pushbutton-type filling valves shall be mounted on the control panel. Force necessary to operate the valve shall not exceed 10 pounds. Actuation of the valve shall simultaneously open the hydraulic circuit and start the fill motor. When in the release position, the valve shall not permit leakage from the aircraft hydraulic system to exceed 10 drops in 5 minutes. Valve and tubing size shall be not less than 3/8-inch tube size.

3.7.13.3 Fill-system relief valves. Two variable relief valves as shown on Figure 1 shall be mounted on the control panel. They shall be adjustable throughout a pressure range of from 0 to 140 pounds per square inch and capable of relieving the total volume discharged by the fill pump. The valves shall have a locking device so that vibrations cannot change the pressure setting. The relief valves shall be so designed that maximum pressure setting will be 140 ±5 pounds per square inch.

3.7.13.4 Fill-system check valve. Two check valves dimensionally interchangeable with AN6247-6 or larger shall be located in the fill system as shown on Figure 1 to prevent reverse flow to the fill system pump.

3.7.13.5 Fill-system filter. A filter shall be provided in the fill system as shown on Figure 1. The filter shall be an AN6235-4A filter element in a filter conforming to or dimensionally interchangeable with MS28720-6 suitable for 145 pounds per square inch operating pressure.

3.7.13.6 Fill-system connections. All lines, tubing, hoses, and connections used in the fill system shall be 3/8-inch tubing size, or larger.

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3.7.14 Pressure check valves. Two pressure check valves that are dimensionally interchangeable with AN6249-16 valves, but suitable for operation at 5,000 pounds per square inch, shall be located as near the high-pressure pump outlet as practical as shown on Figure 1 to prevent reverse flow to the high-pressure pumps. Reverse flow shall not exceed 2 drops per minute at any pressure from 100 to 5,000 pounds per square inch.

3.7.15 Flowmeters. Variable-orifice-type visual flowmeters rated at 300 pounds per square inch gage shall be provided in the return lines as shown on Figure 1. The flowmeters shall be accurate within ±2 percent of full-scale reading. The flowmeters shall be lighted from behind so fluid contamination or air is clearly visible.

3.7.16 Gages.

3.7.16.1 Hydraulic gages. All hydraulic gages shall conform to GG-G-76 except, where conflicts exist, this specification shall govern. All hydraulic gages shall be of the recess type with a mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 5/8 inch beyond the external surface of the panel. The gages shall be mounted by screws and tapped panel holes and shall be connected with flexible AN or MS hose, or commercial equivalent, to permit removal and installation by one man from in front of the panel. The high-pressure gage lines shall be provided with needle shut-off valves and gage test connections to permit calibration of the gages in the panel. Means shall be provided to restore the accuracy of the high and low-pressure gages without removing the gage from the panel. All gages shall be designed for operation at -40° to +180°F. The hydraulic gages shall be corrosion resistant and shall incorporate good quality stainless-steel movements and steel alloy Bourdon tubes. The dial and pointer shall be separated from the gage movement by a wall which will prevent rupture through the front of the gage in the event of gage failure. The gage cases shall incorporate a safety blowout device which will automatically relieve the case pressure should it reach 1/2 pounds per square inch.

3.7.16.1.1 High-pressure gages. High-pressure gages with a dial diameter of 6 inches and a dial range of 0 to 6,000 pounds per square inch shall be provided. The gages shall be accurate to within ±1/2 percent or less of full scale at all operating temperatures specified herein. The high pressure gages shall be protected by accumulator type hydraulic dampers.

3.7.16.1.2 Low-pressure gages. Gages having a dial diameter of 4-1/2 inches or larger and a dial range of 30 inches mercury vacuum to 300 pounds per square inch pressure shall be provided. The gages shall be accurate to within 1 percent of full scale range.

3.7.16.2 Shutoff valves. Needle shutoff valves for the high pressure gages shall be of the panel-mounted type with corrosion-resistant, steel stems. The valves shall be capable of withstanding pressures up to 5,000 pounds per square inch.

3.7.16.3 Gage test fittings. Gage test fittings shall be provided for the high- and low-pressure gages. The following fittings shall be mounted on the control panel adjacent to their respective gages:

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a. AN832-4D nipple

b. AN924-4D nut

c. AN929-4 cap

3.7.16.4 Gage selector valves. Two shutoff-type selector valves shall be mounted on the instrument panel and shall permit pressure selection at the various points as shown on Figure 1. One position of the selector shall isolate the low pressure gages for calibration.

3.7.16.5 Temperature gages. Flush-mounted, remote-reading temperature gages shall be provided. The range of the gages shall be approximately -20° to +200°F and shall indicate within ±5°F of the temperature of the fluid supply to the high-pressure pumps. The gages shall have a 3-inch diameter dial, or larger. The dial faces shall be marked with a red sector as a warning at 160°F and higher. The accuracy of the gages shall not be affected by continued exposure to the climatic extremes specified herein.

3.7.16.6 Fill-pressure gage. A corrosion-resistant gage with a 2-1/2 inch dial face shall be mounted on the test stand panel to indicate the pressure in the fill system. The gage shall have a range of from 0 to 160 pounds per square inch.

* 3.7.17 Thermoswitches. Adjustable-type preset thermoswitches shall sound a warning horn when the oil inlet temperatures to either high pressure pump exceed +165°F. The horn shall be audible for a distance of 50 feet from the test stand when the test stand is operating at full capacity. The thermoswitches shall have an adjustable range of from +150°F to +200°F.

* 3.7.18 Hose assemblies. The hose assemblies specified in Table I shall be provided with each test stand and shall be equipped as specified.

3.7.18.1 All dust plugs and caps shall be attached to the test stand or hose assemblies by metal chains to prevent loss.

3.7.18.2 Adapters. The following adapaters shall be furnished in the stowage compartment:

3.7.18.2.1 Two each coupling halves, Aeroquip part No. 015519-S2-16D and No. T150-S5-12D or equal to and interchangeable with the Aeroquip parts, shall be fastened into two double-threaded couplings similar to AN910 with AND10050-16 threads in one end and AND10050-12 threads in the other. The threaded coupling shall be not more than 3 inches long and shall be capable of withstanding pressures of 5,000 pounds per square inch and proof pressures of 7,500 pounds per square inch.

3.7.18.2.2 Two each coupling halves, Aeroquip part No. 014519-S2-24D and No. 145-S5-16D or equal to and interchangeable with the Aeroquip parts, shall be fastened into two double-threaded couplings similar to AN910 with AND10050-24 threads in one end and AND10050-16 threads in the other. The threaded couplings shall be not more than 3 inches long and shall be suitable for working pressures of 200 pounds per square inch and proof pressures of 300 pounds per square inch.

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TABLE I

Hose Assembly Part Number	Quantity	Hose Ends (One or Both)	Equipped With
MS28759-16-3600	2	Both	Coupling half, Aeroquip fitting or equal, Part No. 155-S5-16D Dust plug, Aeroquip fitting or equal, Part No. 155-S9-16D
MS28741-24-3600	2	Both	Coupling half, Aeroquip fitting or equal, Part No. 145-S5-24D Dust plug, Aeroquip fitting or equal, Part No. 145-S9-24D

* Or a part equal to and interchangeable with the Aeroquip part.

3.7.19 Fittings. All fittings and threaded bosses in the hydraulic circuits and components shall conform to AND10050, MS33656, or MS33657. All 5,000 pounds per square inch pressure AN and MS fittings shall be steel. All bulkhead type, high-pressure fittings shall use an AN6289 locknut with a backup gasket conforming to MS28777. All seals used on fittings shall conform to MS28778.

* 3.7.20 Sampling valves. Fluid sampling valves shall be installed immediately downstream of each high pressure filter and shall be adequately lighted for fluid sampling at night.

3.8 Electrical system. All electrical euqipment and wiring shall conform to NFPA Pamphlet 70. All electric motors shall conform to CC-M-636 or CC-M-641, as applicable. Electrical equipment less than 18 inches from the ground shall be explosion-proof for class I, division 2 location in compliance with section 5115 of Pamphlet 70. The electrical system shall be designed for operation on 220-volt, 3-phase, 60-cycle power and shall be designed for conversion to 440-volt, 3-phase, 60-cycle power without replacing any major component. The voltage tolerance shall be ±10 percent and the frequency ±5 percent. The electrical system shall be connected as shown on Figure 4.

3.8.1 Pump motors. Two design B or C, code D or E, ball-bearing, squirrel-cage induction motors of not less than 60 horsepower continuous rating, each conforming to National Electrical Manufacturer's Association (NEMA) Standard MG1, shall be provided for driving the high- and low-pressure pumps of each system. The pump motors shall conform to CC-M-641 and shall be of sufficient rating to operate the test stand continuously under the conditions specified in 4.3.6, at maximum capacity without exceeding the NEMA recommended temperature limits for the class of insulating materials used. The manufacturer's recommended full-load current shall not be exceeded during operation. The motors shall operate at a pressure altitude of 6,000 feet. Means shall be provided to

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reduce starting current surges to 300 amperes when connected for 440-volt operation and to 400 amperes when connected for 220 volt operation. The high-pressure pumps shall be mounted on the respective motor on mountings conforming to AND20006.

3.8.2 Pump motor control. The test stand shall be provided with a remote-operated magnetic motor control system in accordance with Figure 4, providing motor overload protection and circuit protection. The remote-operated control shall be powered by 110 volts, obtained by reducing input voltage with a transformer. Voltages greater than 120 volts shall not be used on the control panel.

3.8.2.1 Starter and starter case. A starter of sufficient capacity to withstand openings under starting surges shall be provided for each system. A starter case of sufficient size to enclose all remote controls (contactors, overload devices, transformers, starters) for all electric motors shall be furnished. For motor protection, each starter shall have two overload relays which can be manually reset without opening the starter case. A 240/480-volts to 120-volts control transformer shall be provided in the starter enclosure.

3.8.2.2 Pushbuttons. A start-stop, momentary-contact, low-voltage-protection pushbutton shall be mounted on the instrument panel of each system. The push-buttons shall be located and grouped with the hydraulic controls of the respective system. The start buttons shall incorporate a cutout for the boost pump interlock system specified in 3.7.1.7.

3.8.2.3 Circuit breaker. A circuit breaker rated in accordance with NFPA Pamphlet 70 shall be provided for each circuit or circuit branch.

* 3.8.2.4 Wiring. All wiring between electrical components shall be in accordance with NFPA Pamphlet 70. A color code or number code shall be used on all wires and terminal lugs to simplify tracing and reworking of the circuits. Bundling and grouping of the wiring shall conform to paragraphs of MIL-W-5088 titled:

a. Cable bundles

b. Groups

c. Power systems

* 3.8.2.5 Electrical cable and connector. The alternating-current input cable shall be 50 feet long, extra flexible, 6-conductor cable in accordance with MIL-C-5756. The cable shall use three conductors of American Wire Gage (AWG) No. 000 connected to contacts L1, L2, and L3; one AWG No. 4 conductor connector to contact LO (ground); and two AWG No. 12 conductors connected to contact C1 and C2. The cable shall be securely attached to the test stand to remove all the strain from the terminals. The male connector shall be watertight, polarized type conforming to MS27265-1.

* 3.8.2.6 Reverse rotation protection. A mechanical or electrical device shall be provided to prevent operation of the pump motor in the opposite direction of rotation from that required for the pumps, in the event of incorrect phasing.

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3.9 Pneumatic system. A compressed air system shall be provided for pressurizing aircraft reservoirs up to 150 pounds per square inch gage or the unit reservoir up to 50 pounds per square inch gage.

3.9.1 Air compressor. An air compressor delivering 3 cubic feet per minute at 150 pounds per square inch continuously when supplied with air at 77°F and 29.92 inches mercury absolute shall be provided.

3.9.2 Unloader valve. A pilot actuated, unloader valve shall sense pressure at the air receiver and unload the compressor when the receiver is fully charged.

3.9.3 Pressure regulators. Two pressure regulators operating at -40° to +125°F ambient temperature shall be mounted on the control panel. The pressure regulators shall control and limit the pressure to the outlet at any setting of from 5 to 150 pounds per square inch gage and the unit reservoir pressure from 5 to 50 pounds per square inch gage and shall maintain these set pressures at any flow condition including zero flow.

3.9.4 Air pressure gages. Two 2-inch minimum diameter pressure gages, having a scale range of 0 to 200 pounds per square inch shall be included to indicate pressure in the aircraft reservoir and the test stand reservoir. The gages shall be accurate to within 5 percent of full scale range.

3.9.5 Relief valves. Fixed relief valves set at 160 and 60 pounds per square inch gage shall be installed in the system to protect the outlet and reservoir pressurization systems.

3.9.6 Reservoir pressurization valve. A three-way, two-position valve shall be installed between the pressure regulator and the test stand reservoir. In one position the reservoir shall be pressurized and in the other position the reservoir shall be exhausted and the presure regulating outlet blocked off.

3.9.7 Air receiver. An air receiver with minimum capacity of 0.25 cubic feet shall be provided. The air receiver shall have a minimum working pressure of 150 pounds per square inch gage and proof pressure of 300 pounds per square inch gage. The air receiver shall be provided with a drain and sufficient clearance to drain collected water to the exterior of the test stand.

3.9.8 Air filter. A filter conforming to or dimensionally interchangeable with MS28720-12, suitable for 150 pounds per square inch pressure, shall be provided in the air system. The filter shall contain an AN6235-4A element.

* 3.9.9 Air outlet. An AN832-4D nipple, AN924-4D nut, and an AN929-4 cap shall be provided for the outlet connection. The outlet connection shall be located adjacent to the hydraulic outlets. A shutoff valve shall be provided on the control panel to shut off air to the outlet connection. Two 20-foot lengths of AN6270-4-2400 hose shall be provided.

* 3.9.10 Compressor motor. A 220/440-volt, 3-phase, 60-cycle motor shall be provided. It shall be of sufficient rating to operate the compressor at maximum rated pressure and output continuously without exceeding safe temperature limits for the class of insulation used. The compressor motor shall be connected as shown on Figure 2.

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3.9.11 Compressor motor controller. A manual or magnetic push-button, 220/440-volt, 3-phase, 60-cycle controller with two overload relays shall be provided for starting, stopping, and protecting the compressor motor. The compressor control shall be mounted on the control panel in an instrument cluster of all pneumatic controls and gages.

* 3.10 Fan or blower. A fan or blower driven by a separate electric motor shall be furnished. The fan or blower shall provide sufficient cooling air to the air-to-oil cooler to meet all oil cooling requirements. The fan or blower shall normally function whenever the pumps are operating; however, for low-temperature operation, provisions shall be made to disconnect the drive mechanism from the fan or blower. Provisions for accomplishing the requirements specified herein shall be accessible but shall not be on the control panel.

* 3.11 Trailer and housing. The components of the test stand shall be assembled and mounted on a trailer and covered with a weather-resistant housing. The test stand shall meet the requirements of MIL-M-8090 for type II, group C mobility and shall meet the air transportability requirements of MIL-A-8421 in general class aircraft.

3.11.1 Trailer. The trailer shall be of the 4-wheel, 2-wheel-steerable type.

3.11.1.1 Frame. The trailer frame shall be of welded-steel construction. The frame shall be not less than 54 inches nor more than 66 inches wide and shall be not less than 84 inches nor more than 108 inches long. The ratio of width divided by length shall not exceed three-fourths.

3.11.1.2 Springs. The springs on the trailer shall provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be of such construction that not more than 3 inches is added to the frame height for spring deflection.

* 3.11.1.3 Parking brakes. Mechanical parking brakes in accordance with MIL-M-8090 shall be provided. Brakes for both rear wheels shall be interconnected and shall operate from one handle located near the tow bar.

3.11.1.4 Reflectors. Reflectors required by MIL-M-8090 need not be provided.

* 3.11.1.5 Lifting and tiedown fittings. Lifting and tiedown fittings conforming to Drawing 48B7796 or 49B6564 shall be provided and shall be capable of withstanding the loadings specified in MIL-A-8421. The unit shall be designed for forklifting. The fork lifting area will be protected with a "U" type channel to prevent fork damage and slipping of the stand from the forks.

* 3.11.1.6 Pintle hook. A pintle hook in accordance with MIL-M-8090 and conforming to MS51335 shall be mounted on the center of the rear frame.

* 3.11.2 Control panel. A control panel (or panels) shall be provided for mounting of all meters, valves, indicators, gages, switches and controls furnished with the test stand. The panel(s) shall be constructed of No. 12 US gage (0.1046 inches), or thicker, sheet steel, and shock mounted from the welded steel, self supporting frame work. In turn all meters, valves, indicators, gages, switches and controls subject to error from vibration shall be shock mounted from the control panel(s).

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* 3.11.2.1 Grouping and color coding of controls and gages. Each system shall have the fill button, fill system relief valves, bleed button, and sight tube grouped together. The pneumatic system controls and gages shall be grouped together on one section of the control panel.

* 3.11.2.1.1 No. 1 system. All controls and gages for the No. 1 system shall be grouped in one section of the control panel. Valve packing nuts, handles, buttons, and gage face rings shall be painted red.

* 3.11.2.1.2 No. 2 system. All controls and gages for the No. 2 system shall be grouped in one section of the control panel. Valve packing nuts, handles, buttons, and gage face rings shall be painted green.

* 3.11.2.1.3 Warning lights. The filter warning lights shall be grouped together in a conspicuous section of the panel. All warning lights shall be of the press-to-test type conforming to MS25331.

* 3.11.2.1.4 Color coding of panel. In lieu of color coding each nut, handle, button, and gage face ring as specified in 3.11.2.1.1 and 3.11.2.1.2, the control panel proper may be color coded in order to distinguish the two systems.

* 3.11.2.1.5 Illumination. The operator control panel shall be provided with sufficient illumination to permit night operation.

* 3.11.2.2 Location and position. The control panel shall be located at a sufficient height above the ground and inclined at a sufficient angle (10 degrees minimum) to facilitate convenient observation and operation by an operator approximately 5 feet 10 inches tall. The control panel frame shall be bolted or welded to the trailer bed and so positioned that the trailer wheels, tow bars, et cetera, will not interfere with the operator.

3.11.3 Housing. The test stand components and control panel shall be completely enclosed and protected by a weather-resistant housing. The housing shall consist of a welded-steel framework covered with No. 16 US gage (0.0598 inch) sheet steel, or heavier, with doors and adjustable louvers as required. The housing shall be adequately braced to prevent distortion or excessive vibration during handling and operation of the test stand or handling of the housing while removed from the test stand. The test stand shall be so designed that water or hydraulic fluid spilled within the housing will drain to the exterior and will not collect within the enclosure.

* 3.11.3.1 Hooks. Metal hooks shall be provided on the exterior of the housing to permit temporary storage of the hydraulic hoses. Additional hooks shall be provided on the exterior of the housing to permit temporary storage of the power cable. All these hooks shall be designed to permit flat positioning against the housing when not in use. The hooks shall have adequate bracing and be of sufficient strength to support all hoses, filled with oil, and the related hardware during all of the tests. The hooks and their attaching members shall not bend or deflect during specified tests.

* 3.11.3.2 Internal accessibility and ventilation. The housing shall have sufficient access and ventilating doors and adjustable louvers to permit access for operating, servicing, and ventilating the test stand. The doors and louvers shall be free from binding and shall be easy to operate with one hand. Provisions shall be incorporated to secure the doors rigidly in the full-opened or full-closed position. Direction of rotation for all fasteners or latches shall be marked clearly.

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* 3.11.3.3 The housing shall be easily removable by actuating not more than 10 quick-acting disconnecting fasteners and lifting or sliding the entire housing off the remainder of the stand assembly. Handles shall be provided for this purpose and lifting eyes for use with hoists. These eyes must fold flat when not in use. The housing does not have to be a one piece design but can consist of removable panels and braces; however, when this method is used, internal components or pieces will not be secured to the braces or housing. Direction of rotation for all fasteners or latches shall be clearly marked.

* 3.11.3.4 The housing shall be capable of supporting a 200-pound weight distributed evenly over an area 4.5 by 5 inches and applied in any location on the top of the test stand. There shall be no deflection or distortion caused by this weight.

* 3.11.4 Compartments. A handbook compartment of 2 by 9 by 12 inches, or larger, and an enclosed accessory box of 8 by 8 by 24 inches, or larger, shall be permanently and securely attached at an accessible point within the housing. Provisions shall be made for storage of the hose assemblies and accessories specified herein within the outline of the housing. The compartments may be constructed on the trailer bed or may be fastened to the housing assembly.

* 3.12 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

* 3.13 Electromagnetic interference. The test stand shall meet the requirements of MIL-STD-826.

3.14 Dimensions. The length and width of the test stand and housing shall be not less than the trailer frame assembly nor extend beyond the trailer wheels at any point. The overall height of the test stand shall be held at a minimum consistent with good design practices, but shall not exceed 44 inches above the top of the tires.

3.15 Weight. The dry weight of the test stand, including the trailer assembly, shall not exceed 4,500 pounds (without hydraulic fluid, et cetera).

3.16 Finishes and protective coatings.

3.16.1 Cleaning, painting, and plating. Cleaning, painting, and plating shall be in accordance with MIL-STD-808.

3.16.1.1 Tubing and components that have been treated with corrosion-resistant coatings need not be painted.

3.16.2 Color. Color shall be in accordance with MIL-STD-808, exterior Code No. DG, interior Code No. LG.

* 3.16.2.1 Color coding of hoses. Attach metal strap or tape at each end of the hoses. Red shall be for primary and green for secondary.

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3.17 Markings.

3.17.1 A caution plate containing the following information shall be permanently mounted on the instrument panel adjacent to the selector valves:

DO NOT OPERATE WHEN THE FLOW CONTROL VALVE IS OPEN

3.17.2 An engraved or etched, plastic or metal nameplate containing the following information shall be permanently attached to each pressure connection:

PRESSURE RETURN NO. _____

3.17.3 An engraved or etched, plastic or metal nameplate containing the following information shall be permanently attached to each suction connection:

SUCTION RETURN NO. _____

3.17.4 A permanently engraved plate containing low temperature adjustment directions for the oil coolers shall be provided on the control panel.

3.17.5 An etched or engraved, metal or plastic nameplate containing instructions as to the location, operation, and function of the manifold valve shall be permanently attached to the control panel.

3.17.6 An engraved or etched, permanent-type correction chart shall be provided near the flowmeters.

3.17.7 A permanent-type nameplate containing complete instructions for converting the test stand from operation on 220 to 440 volts, or vice versa, shall be fastened inside the starter case specified in 3.8.2.1.

3.17.8 The needle shutoff valves shall have the manufacturer's cold-working pressure rating stamped on the valve body with an arrow indicating the flow direction.

3.17.9 The trailer parking brake control shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be 1/4 inch black letters and shall be visible to the operator.

3.17.10 All items mounted on the control panel shall be identified as to use and cautions by permanently etched or engraved, metal or plastic plates attached to the panel. Directions for increasing and decreasing flow shall be indicated for all controls.

3.17.11 An 8- by 10-inch, engraved, permanent-type diagram plate, arranged similar to Figure 1 to show the hydraulic circuit of the test stand, shall be securely fastened to the center of the instrument panel.

3.17.12 The following information shall be stenciled in 1-inch black letters on each of the four sides of the stand:

FOR AIR SHIPMENT THIS TEST STAND IS DESIGNED FOR 8 G FORE AND AFT,
3 G SIDEWARD.

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3.17.13 An etched or engraved, permanent-type caution plate shall be securely attached adjacent to the high-pressure gage on the control panel and shall read as follows:

CAUTION - REDUCE FLOW TO 15 GPM (EACH SYSTEM) OR LESS FOR OPERATION ABOVE 3,000 PSI AND REPLACE 3,000-PSI HOSES WITH 5,000-PSI HOSE OR TUBING FOR OPERATION ABOVE 3,000 PSI.

3.17.14 Drawbar or housing marking. The following information shall be marked in 1-inch black letters on the drawbar or front of the inclosure:

MAX SPEED 20 MPH.

3.17.15 The exterior of the test stand shall be stenciled to indicate the pressure, center-of-gravity, and other markings necessary for proper handling and maintenance.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following additional marking shall be included:

RATING: 0 TO 27 GPM TO 3,000 PSI EACH SYSTEM, 0 TO 54 TOTAL.
15 GPM TO 5,000 PSI EACH SYSTEM, 0 TO 30 TOTAL.

TOWING SPEED: 20 MPH MAX.

3.19 Workmanship. The test stand, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, bolts, et cetera.

3.19.1 Riveting. Riveting operations shall be performed to insure tight and satisfactory heading of rivets.

3.19.2 Cleaning. The test stand shall be thoroughly cleaned. Loose, spattered, or excess solder; metal chips; and other foreign materials shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements..

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4.2 Classification of tests. The inspection and testing of the test stand shall be classified as follows:

a.	Preproduction testing	See 4.4
b.	Acceptance tests	See 4.5

4.3 Test conditions.

* 4.3.1 Standard atmospheric conditions. Unless otherwise specified, tests shall be conducted at atmospheric pressure (approximately 29.92 inches mercury) and at room temperature (approximately 70°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Test period. Unless otherwise specified herein, any test period shorter than 1 hour shall not be credited as test time.

4.3.3 Apparatus and instruments. Apparatus and instruments used in conjunction with testing specified herein shall be of laboratory precision-type, shall be checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation, and shall be calibrated in accordance with MIL-C-45662.

4.3.4 Fluid flow. The calibration and performance checks of the displaced fluid shall be determined by means of a calibrated flowmeter with a certified accuracy of 1 percent or better of reading.

4.3.5 Preliminary run-in and calibration. Any preliminary run-in of the test stand as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the test stand for preproduction or acceptance tests. All instruments shall be calibrated and shall conform to the accuracies specified herein. A certified log of all preliminary running time, calibration data, et cetera, shall be submitted to the procuring activity.

4.3.6 Pump tests. The following tests shall be performed on one new high-pressure and one new low-pressure pump to establish the 1,000-hour life expectancy.

* 4.3.6.1 Efficiency test. The high pressure pump shall be connected to a calibrated power source and operated at 3750 revolutions per minute to deliver 30 gallons per minute flow at 3000 pounds per square inch while pumping hydraulic fluid conforming to MIL-H-5606 at 68°F. Not more than 60 horsepower shall be required to drive the high pressure pump.

4.3.6.2 Pump endurance. The high-pressure pump shall be driven at 3,750 revolutions per minute for a minimum of 1,000 hours under the following conditions. Fluid temperatures during the 1,000 hours shall be maintained at 80°F 25 percent of the time, 120°F 50 percent of the time and 160°F the remaining 25 percent of the time for each flow condition. Inlet pressure shall be maintained at the minimum recommended by the pump manufacturer for 10 percent of the time, normal recommended pressure 80 percent of the time, and 300 pounds per square inch gage 10 percent of the time for each flow condition. Output flow conditions shall be as follows:

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Length of run (Hours)	Output Pressure (Pounds per square inch gage)	Output Flow (Gallons per minute)
400	3000	30
200	5000	15
200	3000	10
50	750	20
50	1500	5
100	3000	30

By use of a quick-acting (0.1 second) solenoid valve in the pressure line, the pump shall be compensated from the preset flow to zero flow and vice versa once each minute during this test. The pump case drain flow reading shall be recorded.

4.3.6.3 Compensation time. Inlet pressure, outlet pressure, and delivery of the pump shall be recorded at 15 gallons per minute and 5,000 pounds per square inch, 30 gallons per minute at 3,000 pounds per square inch, and 20 gallons per minute at 1,000 pounds per square inch to show compensating characteristics of the pump. The pump shall compensate from zero flow to full flow and vice versa in 0.05 second and the pressure shall not exceed 133 percent of the selected pressure at any time.

4.3.6.4 Low-pressure pump tests. The low pressure pump shall be tested at rated flow and speed for 1,000 hours under the following conditions:

Running Time (Hours	Inlet Conditions (Pounds per square inch gage or Inches Mercury)	Outlet Pressure (Pounds per square inch gage)
300	125 pounds per square inch gage	250
200	30 pounds per square inch gage	155
300	15 inches mercury	95
100	10 inches mercury to 0 pounds per square inch gage	110
100	125 pounds per square inch gage	250

4.4 Preproduction testing.

4.4.1 Preproduction test sample. One test stand shall be subjected to the preproduction tests specified in 4.4.3.

* 4.4.2 Preproduction test report. After completion of the preproduction tests, a preproduction test report shall be prepared in accordance with MIL-STD-831 and a complete copy furnished the procuring activity.

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* 4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the above report (same quantity and type of copies) or shall be included as part of that report.

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that if modified should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. Indication of the manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

g. Total time of preproduction testing shall be divided by the total number of failures encountered during these tests. The result shall be 152 times 1-1/2 (228 hours) or more. Should the MTBF be less than 228 hours, then testing shall continue after maintenance until ability to meet this time is proven. For these tests, "failure" shall be defined as an inability to operate within specification limits that are not correctable by normal routine maintenance such as filter replacement.

4.4.3 Preproduction tests. Preproduction tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests

b. Sampling tests

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4.5.1 Individual tests. Each test stand shall consist of the following tests as described under 4.6:

a. Examination of product (see 4.6.1)

b. Flow test (see 4.6.2)

c. Cycling test (see 4.6.3)

d. Fluid temperature test (see 4.6.4)

4.5.2 Sampling tests

4.5.2.1 Lot. All test stands offered for delivery at one time shall be considered a lot for purposes of inspection.

* 4.5.2.1.1 Sampling for test. One out of 10 stands shall be selected at random and subjected to tests. (Note: This item also shall undergo all other specified tests.)

4.5.2.2 Tests. Each sample test stand selected in accordance with 4.5.2.1.1 shall be subjected to the following tests as described under 4.6:

a. Semi-endurance test (see 4.6.5)

b. Examination (see 4.6.6)

4.6 Test methods.

4.6.1 Examination of product. The test stand shall be inspected to determine compliance with requirements specified herein with respect to parts, accessibility, ease of maintenance, materials, workmanship, and markings.

4.6.1.1 Leakage of valves. The filter bleed valves and high-pressure check valves shall be tested for leakage. The filter bleed valve leakage shall not exceed 2 drops in 5 minutes at a differential pressure of 275 pounds per square inch; the fill system valve leakage shall not exceed 10 drops in 5 minutes at 150 pounds per square inch differential pressure; and the high-pressure check valve leakage shall not exceed 2 drops per minute at 5,000 pounds per square inch differential pressure.

4.6.2 Flow tests. The pressure hose for each system shall be connected through an external 1-inch size, or larger, high-pressure relief valve and back into the return hose of the respective system. The reservoir selector valves shall be positioned in the stand reservoir position. The reservoir may be pressurized. Each pump volume stop shall be set to the selected flow with the bypass valve full open. The zero flow pressures shall then be set with the bypass valve closed. The external relief valves shall be adjusted to obtain the selected flow and pressure. All data recorded shall be at the steady-state condition. Data shall be recorded for the following settings:

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Volume Stop Setting (Gallons per Minute)	Zero Flow Pressure Setting (Pounds per Square Inch)	Approximate Full Flow Pressure (Pounds per Square Inch)
27	3,000	2,700
27	2,000	1,800
27	1,000	800
27	500	400
20	3,000	2,800
20	1,500	1,350
15	5,000	4,700
15	3,000	2,800
15	1,500	1,350
10	5,000	4,800
10	3,000	2,800
10	1,500	1,400
5	5,000	4,800
5	3,000	2,800
5	500	400

4.6.3 Cycling test. For this test, a quick-acting (0.2 second or less) shutoff valve shall be installed between the pressure outlet connection and the relief valve specified in 4.6.2. The pressure and flow settings listed in 4.6.2 shall be repeated while cycling the shutoff valve to on and off positions at least 5 times at each setting. Steady-state pressure and flow readings shall be recorded for both the flow and no-flow conditions. The test stand shall then be operated at 15 gallons per minute flow and 5,000 pounds per square inch pressure from each system for 1 hour and inspected for leakage of fluid.

4.6.4 Fluid temperature test. The temperature of the fluid entering the high-pressure pumps on the test stand shall be raised above 160°F and the thermoswitches shall sound the warning horn or bell before the fluid temperature exceeds 165°F but not before it reaches 160°F. Each system shall be checked independently. The warning horn or bell shall be audible in a radius of 50 feet from the test stand in any direction when the test stand is running at full load. On completion of this test, all filter elements in the unit shall be replaced with new elements.

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4.6.5 Semi-endurance test. The test stand shall be subjected to 20 percent of each test specified in 4.6.9 (a through e).

4.6.6 Examination. At the completion of the test specified in 4.6.5, the test stand shall satisfactorily pass the tests specified in 4.6.2 and 4.6.3.

4.6.7 Initial disassembly inspection. The test stand shall be disassembled to the maximum practical extent and completely inspected, the breakdown weights determined, and measurements made which may be used to determine wear. Disassembled to the maximum practical extent is defined as disassembly of all moving contacting surfaces which have been assembled by the manufacturer of the test stand and which are subject to wear (gears, bearings, control mechanisms, et cetera). In addition to manufacturer assembled items, the high-pressure pumps shall be disassembled. All gages, meters, and instruments shall be laboratory checked against certified calibrating equipment, and failure to meet the accuracy requirements specified herein, for any item, shall be cause for rejection. All requirements which may be determined by weight or dimensional measurements and inspection shall be checked for compliance, and failure to meet any of these requirements shall be cause for rejection. Following this inspection, the test stand shall be reassembled and subjected to the remainder of the preproduction tests.

4.6.8 Environmental tests. The test stand shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-STD-810. Unless otherwise specified, upon completion of each test, the test stand shall be subjected to the tests specified in 4.6.2 and 4.6.3. Failure of the test stand to operate as required during or after each exposure, failure of any item or assembly due to deterioration from exposures in any test, or formation of any undesirable condition that cannot be remedied by routine maintenance shall be cause for rejection.

4.6.8.1 Low temperature. The test stand shall be subjected to low temperature in accordance with method 502, except the storage temperature shall be -65°F. The test circuits shall include two MS28759-16-1200 pressure hoses, relief valves to build up 3,000 pounds per square inch pressure, and two MS-28741-24-1200 return lines. The entire test circuit shall be contained within the cold chamber and the test stand shall not be started until after the temperature of the stand and fluid have stabilized at -40°F. The fluid in the reservoir shall not be heated prior to starting or during the test from any external source. After starting, the flow and pressure of both systems shall be slowly increased to the rated output of 3,000 pounds per square inch. The test stand shall attain rated output within 30 minutes after starting and shall be operated at rated output for 10 minutes. The test required by 4.6.2 and 4.6.3 shall be performed at -40°F. All controls, gages, meters, valves, systems, and accessories shall be checked for satisfactory operation in the -40°F ambient temperature. Low-temperature instructions as contained on the test stand instrument panel shall be followed. The high-pressure gage shall be accurate to within 1/2 percent of full scale. The system and fluid shall be stabilized at -20°F and the output of the fill system shall supply 2 gallons per minute flow at 150 pounds per square inch gage pressure with a fluid temperature of +20°F.

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4.6.8.2 High temperature. The test stand shall be subjected to high temperature in accordance with method 501. Upon completion of the exposure to high temperatures, the ambient temperature shall be lowered to 125°F. The tests in 4.6.2 and 4.6.3 shall be performed at 125°F. An external cooler may be used to maintain the temperature of the fluid between 140° and 160°F. The high-pressure gage shall be accurate to within ±1/2 percent of full scale at all settings. The flowmeter shall be accurate within ±1/2 gallons per minute of actual flow. Both system pumps shall be adjusted to 27 gallons per minute with the high-pressure relief valve in each system adjusted to 750 pounds per square inch gage (routing full flow through the cooler). The unit shall continue to run at this flow and pressure setting until the fluid temperature stabilizes and remains stable for a minimum of 15 minutes. No external circuit cooler or reservoir shall be used for this portion of the test. The test stand cooler shall maintain the fluid temperature below 160°F. After this test, the test stand shall be inspected for leakage of fluid.

4.6.8.3 Fungus resistance. The test stand shall be subjected to fungus in accordance with method 508, procedure I.

4.6.8.4 Sand and dust. The test stand shall be subjected to sand and dust in accordance with method 510.

4.6.8.5 Humidity. The test stand shall be subjected to humidity in accordance with method 507.

4.6.8.6 Salt spray. The test stand shall be subjected to salt spray in accordance with method 509.

4.6.8.7 Rain. The test stand shall be positioned on the test chamber floor in its normal operating position. After exposure to the rain test for 1 hour and while still in the rain test, the test stand shall be started and both systems operated through the tests of 4.6.2 and 4.6.3. Failure of the test stand to start or operate satisfactorily during the rain test shall be cause for rejection. Upon completion of the rain test, all surfaces that might collect moisture shall be inspected to determine adequacy of drainage facilities and any collection of water in excess of 1/8 inch deep or which could be detrimental to the performance of the test stand shall be cause for rejection.

* 4.6.9 Endurance test. Both systems of the test stand shall be connected in series to a test rig through the 30-foot pressure hoses and 30-foot suction hoses specified herein. The test rig shall include a reservoir, restrictors to create the prescribed vacuum in the return lines, and a cooler or coolers to maintain fluid temperatures at approximately 130°F. The ambient air temperature shall be maintained at 75°F or higher during the endurance test. Readings shall be recorded every hour of all gages, meters, and instruments on the test stand and test rig. Output flow, fluid temperature, fluid pressure, inlet fluid temperature and pressure, motor voltage, amperage, power factor, and kilowatts shall be recorded each hour. The blower fan and cooler shall be operated continuously during all phases of the endurance and calibration test. (See 4.6.12) Endurance test runs shorter than 1 hour shall not be counted as part of the endurance test. The test stand shall first complete the run-in specified in 4.3.5 and shall then be subjected to the following 350 hours of endurance testing. The fill motor and pump and air compressor shall be operated continuously during this period. During the endurance tests, the flow shall be instantaneously (less than 0.1 second) varied from the preset

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flow to zero flow, by means of a solenoid operated valve, and instantaneously returned to the preset flow at the end of 15 ±2 seconds. This cycle shall be accomplished once each minute during the endurance test. Any damage to the pump, pump compensator, flow indicator, gearing, or other components shall be cause for rejection

a. 100 hours operation at 54 gallons per minute and zero flow pressure setting at 3,000 pounds per square inch. Pressure shall be discharged through the 1-inch outlets and returned through the 1-1/2-inch inlets with return pressures of 125 pounds per square inch gage (reservoir bypassed).

b. 50 hours operation at 15 gallons per minute from each system and zero flow pressure at 5,000 pounds per square inch. Pressure shall be discharged through the 1-inch outlets and returned with pressures of 30 ±2 pounds per square inch gage (reservoir in circuit).

c. 100 hours operation at 54 gallons per minute and zero flow setting at 3,000 pounds per square inch with suction to the low-pressure pump reduced to 15 inches mercury vacuum. Pressure shall be discharged through the No. 1, 1-inch outlet and returned through the No. 2, 1-1/2-inch inlet.

d. 50 hours operation at 10 gallons per minute from each system with zero flow pressure setting at 1,000 pounds per square inch and suction of 10 inches mercury to 0 pounds per square inch gage.

e. 50 hours operation at 10 gallons per minute from each system with zero flow pressure setting at 1,000 pounds per square inch and return pressures of 125 pounds per square inch gage.

4.6.9.1 The test specified in 4.6.2 shall be conducted. These data shall be compared with data recorded in 4.6.9. Results shall indicate that the test stand is in satisfactory condition and that no failure or undue wear has resulted from the endurance test.

4.6.10 Electrical equipment tests. The electrical system shall be inspected to determine compliance with 3.8 and all items shall be shown to have Underwriters' Laboratories approval or listing as required. During the last hour of the full capacity run specified in 4.6.9a, the temperature rise in the motor windings shall be determined by the resistance method and if the temperature rise exceeds the safe temperature limits for the class insulation used, as prescribed by American Standards Association Standard C50, it shall be cause for rejection. The motors shall provide safe temperature rise at 6,000 feet pressure altitude and full load and 104°F ambient.

4.6.10.1 Operation of the thermoswitch shall be checked at least twice during the testing specified in 4.6.9, and shall sound the warning horn or bell when the fluid to the high-pressure pump reaches not less than 160°F, or not more than 165°F. Each system shall be independently checked.

4.6.10.2 The pump motor shall be overloaded to 125 percent of full-load current and sustained until overload relays trip and shut off the motor. The overload relays shall trip before any damage is caused to the motor and shall not take longer than 15 minutes to open the main contactor. Ambient temperature for motor and control shall be 104° ±9°F.

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4.6.10.3 The test stand shall be operated for 2 hours at 27 gallons per minute and zero flow pressure setting at 3,000 pounds per square inch, each system, with phase-to-phase voltage measured at the plug end of the electric cable reduced to 198 to 200 volts. The temperature rise in the motor windings shall be determined by the resistance method and shall not exceed the safe temperature limits for the class of insulation used. Ambient air temperature shall be held at 104°F, or higher, during this test.

4.6.10.4 The test stand shall be reconnected for 440-volt operation as specified in 3.8 and operated for 2 hours in 104°F, or over, ambient temperature. The temperature rise in the motor winding shall then be determined by the resistance method and shall not exceed the safe temperature limits for the class of insulation used.

4.6.10.5 Other motors (fill pump and compressor) shall be tested and shown to meet requirements of 4.6.10, 4.6.10.2, 4.6.10.3, and 4.6.10.4.

4.6.11 Air compressor and fill pump tests. The performance of the compressor and fill pump shall be checked at least once during each preproduction test. Failure of the drive couplings or any marked decrease in performance or failure of a component part of the pressurizing or fill system shall be cause for rejection. The output of the compressor and fill pump shall be determined and shall equal or exceed the requirements specified in 3.7.13.1 and 3.9.1.

4.6.12 Altitude. A test report shall be submitted and shall contain calculations showing that the stand will continuously deliver full-rated capacity while operating at a pressure altitude of 6,000 feet.

* 4.6.13 Electromagnetic interference. The test stand shall be tested as a complete unit for radiated and conducted interference in accordance with MIL-STD-826 for class Ib equipment. The antenna for the radiated test shall be placed 6 feet from the outer perimeter of the unit at the point of maximum radiation.

4.6.14 Noise level. The noise level of the test stand shall be measured at head level (5 feet 8 inches) in a circle of 25-feet radius centered on the geometric center of the unit at 12 positions equally spaced every 30 degrees and at head level at the approximate operator position while the test stand is operating at 54 gallons per minute and 2,700 pounds per square inch. The test stand shall not exceed the noise levels shown below. Measurements shall be taken out-of-doors on a paved surface.

Cycles Per Second	Not Exceeding (decibels) Operator Position	At 25-Feet Radius
37 to 75	106	91
75 to 150	96	81
150 to 300	88	73
300 to 600	85	70
600 to 1,200	84	69
1,200 to 2,400	83	68
2,400 to 4,800	82	67
4,800 to 9,600	81	66

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4.6.15 Valve tests. The flow control valves shall be tested by connecting the test stand outlets to external flowmeters. With the pumps set to operate at 1, 3, 5, 10, 15 gallons per minute at 4,700 pounds per square inch; 1, 3, 5, 10, 20, 27 gallons per minute at 2,700 pounds per square inch; and 1, 3, 5, 10, 20, 27 gallons per minute at 1,000 pounds per square inch, each flow control valve shall be operated to determine that it will hold the outlet flow steady at the desired value. Failure of the valves to regulate the flow to ±1/4 gallons per minute at any setting from 0 to 5 gallons per minute and to ±1 gallon per minute at any setting from 5 to 27 gallons per minute shall be cause for rejection. The flow control valves shall be tested for leakage with the pumps operating at 3,000 pounds per square inch. During these tests, the flow control valve settings shall not be influenced by vibration from the test stand. Any leakage in excess of 1 drop per minute shall be cause for rejection.

* 4.6.16 Test stand operation (tilted position). The complete test stand shall be rested on a platform and shall be kept from rolling by the parking brakes only. The platform supporting the test stand shall be tilted in eight different directions, 45 degrees apart, on an 8-1/2 degree incline with the true horizontal plane. The test stand shall be started and operated at rated output and 2,700 pounds per square inch for 15 minutes in each position. The test stand shall perform satisfactorily in every respect, and the parking brakes shall prevent the test stand from moving on the platform. While tilted in each of the eight positions, 25 gallons of fluid shall be pumped from the reservoir and then replaced to check the capacity specified in 3.7.6

4.6.17 Support test. A 200-pound weight shall be applied slowly and uniformly to an area 4-1/2 by 5 inches in at least six random positions on top of the test stand housing, including a reading in the center of each panel. The test stand shall not be permanently deformed by the application of the weight.

* 4.6.18 Mobility tests. The test stand shall be subjected to and shall pass all tests specified in MIL-M-8090 for type II, Group C, mobility except requirements for service brakes and all related paragraphs and the requirement for bumpers. Upon completion of the mobility tests, the tests specified in 4.6.2 and 4.6.3 shall be repeated. Calibration of gages and minor adjustments will be permissible.

4.6.19 Serviceability. The test stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular note shall be made with regard to provisions to prevent accumulation of dirt, snow, ice, et cetera, which may hinder servicing and operation.

4.6.20 Air transportability tests. The test stand shall be subjected to the tests specified in MIL-A-8421. The test stand shall be secured during this test by means of the lifting and tiedown fittings only. Upon completion of this test, the test stand shall be inspected to determine adequacy of bracing, adequacy of tiedown and lifting fittings installation, and security of mounting of all components. The test stand will not be required to be operable upon completion of the 8-g acceleration loading. If it can be shown by computation that the test stand and all components will withstand the 8-g acceleration loading, the computation may be substituted in lieu of the test.

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4.6.21 Backpressure indicator. The filter backpressure indicators shall be tested for proper operation. The indicators shall be tested with at least five different line pressures, including the highest and lowest operating pressure.

4.6.22 Final disassembly inspection. The test stand shall be disassembled to the maximum practical extent for a thorough examination of all parts to disclose failure, distortion, excessive wear, and other unsatisfactory conditions. Measurements made to determine wear, shall be compared with similar measurements made during initial disassembly inspection. Test stands not conforming to requirements specified herein including the maintainability concept of 3.5.2 shall be rejected.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Each test stand shall be preserved in accordance with MIL-P-116, Method I with P-2 preservative on all exposed bare metal surfaces. Hoses shall be coiled and stored in the hose storage compartment. The ends of electrical cables shall be wrapped with barrier material conforming to MIL-B-121, Grade A, and secured in place with tape conforming to PPP-T-60. The cables shall be wrapped on the cable rack of the unit.

* 5.1.2 Level C. Each test stand shall be preserved and packaged to prevent corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. This may conform to supplier's commercial practice when it meets the requirements of this level.

* 5.2 Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Each test stand preserved as specified in 5.1.1 shall be packed in a MIL-C-104, Type I, Class 2, Style A crate. The towbar shall be removed and attached to the base on the inside of the crate. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Containers shall be of uniform size and shape and of minimum tare and cube.

* 5.2.2 Level B. Each test stand preserved as specified in 5.1.1 shall be shipped as a mobile (uncrated) unit. A shroud conforming to MS27766 shall be fabricated and placed over the unit. The cover shall be securely fastened to the unit with its tie-downs.

* 5.2.3 Level C. Each test stand preserved as specified in 5.1.2 shall be shipped as a mobile unit. All vents and other openings shall be sealed with waterproof tape.

* 5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type MK-3A test stand is intended for use in ground testing and maintenance of aircraft hydraulic systems.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of specification.

b. That the contractor will submit proof that the electrical system specified in 3.8 which will be supplied under this specification meets the requirements of Pamphlet 70. The label or listing of the Underwriters' Laboratories, Incorporated, will be accepted as evidence of conforming to the specification requirements. In lieu of the label or listing, the contractor may submit independent proof satisfactory to the procuring activity that the electrical system conforms to the requirements of Pamphlet 70. Compliance with these requirements does not absolve the contractor from complete compliance with the other requirements of this specification.

c. Levels of preservation, packaging, and packing required (see Section 5).

6.3 The high-pressure pumps on this test stand are mounted on standard accessory drive pads. Aircraft pumps can be substituted by the use of the following parts:

AN4139	Adapter-Type XVI Drive Flange
AN4182	Adapter-Type XVII Drive Spline

6.4 The high-pressure pumps are rated (see 3.7.1) and tested at 3,750 revolutions per minute (see 4.3.6) at 30 gallons per minute, 3,000 pounds per square inch to insure interchangeability with pumps on gasoline-engine-driven test stands.

* 6.4.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \quad \text{where } x = \frac{\text{Mission Time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

* 6.4.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

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* 6.4.3 All MK-3A test stands will have the following:

a. The operator control panel shall be provided with illuminaiton sufficient to permit night operation.

b. Fluid sampling valves shall be installed immediately downstream of each high pressure filter and shall be adequately lighted for fluid sampling at night.

c. All warning lights shall be of the "press-to-test" type conforming to MS25331.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 82

Review Activity:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 4920-F028

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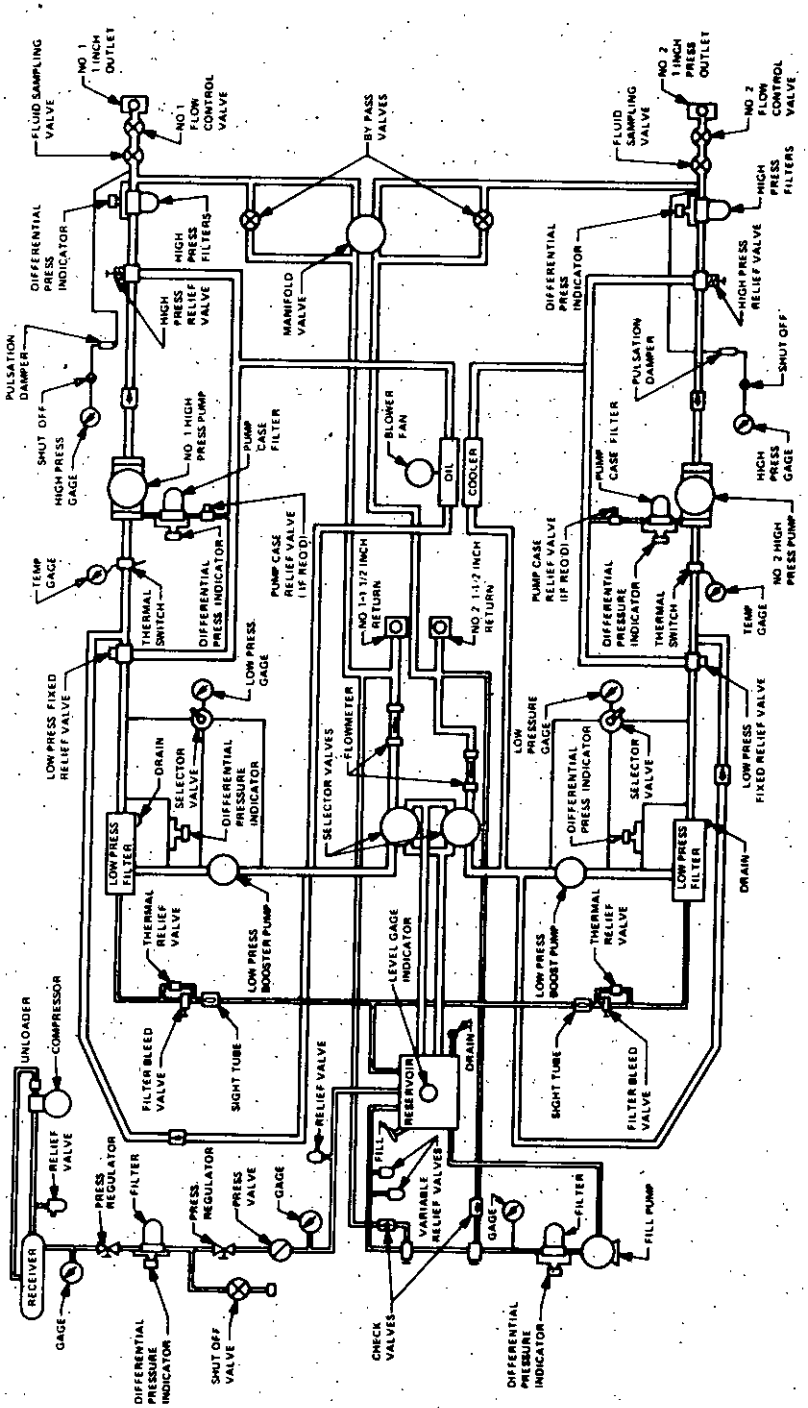


FIGURE 1. SCHEMATIC-HYDRAULIC SYSTEM

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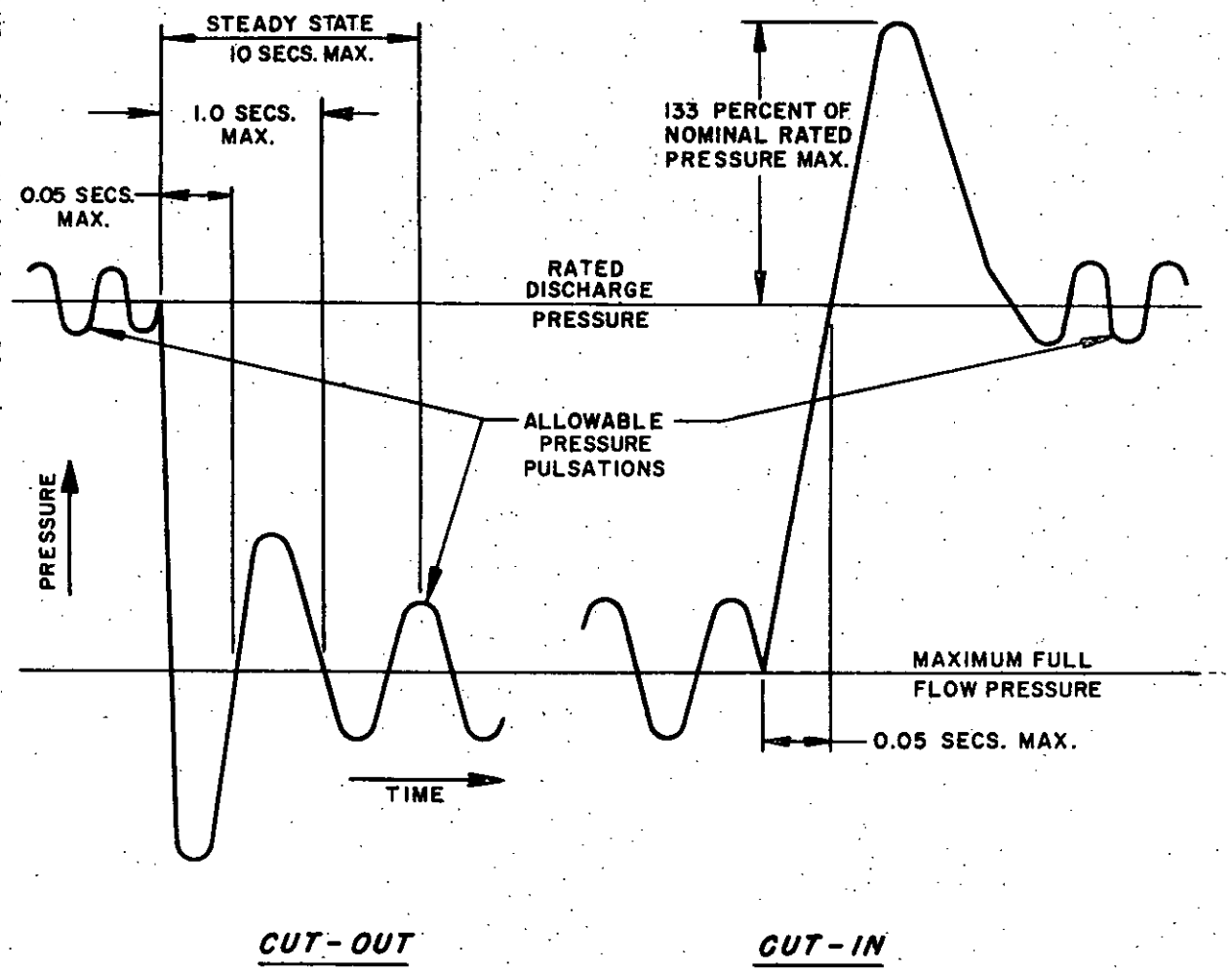


FIGURE 2 TYPICAL PRESSURE VS TIME CHART

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CAPACITY SHALL BE WITHIN THE LIMITS SHOWN WHILE OPERATING AT 3750 R.P.M. & USING HYDRAULIC FLUID AT 140° F.

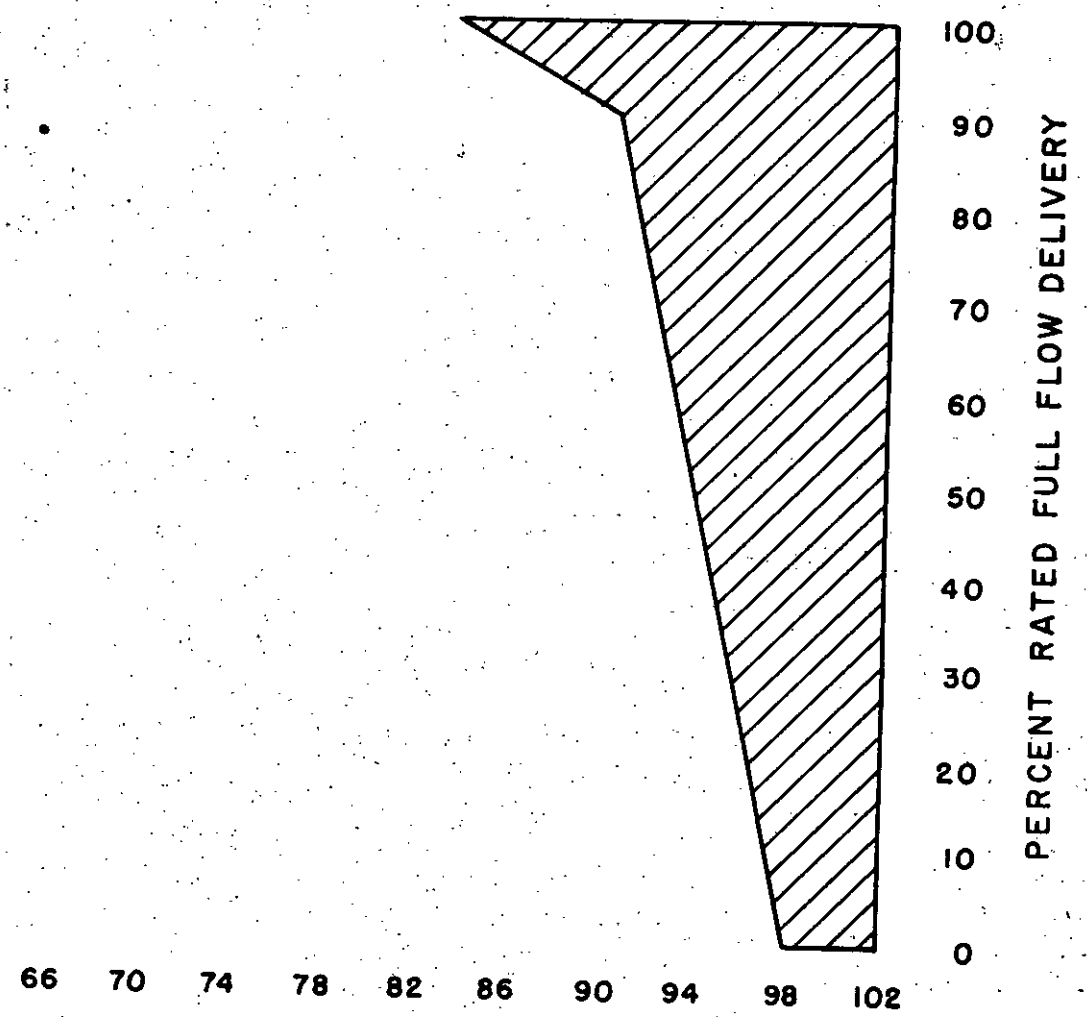


FIGURE 3.

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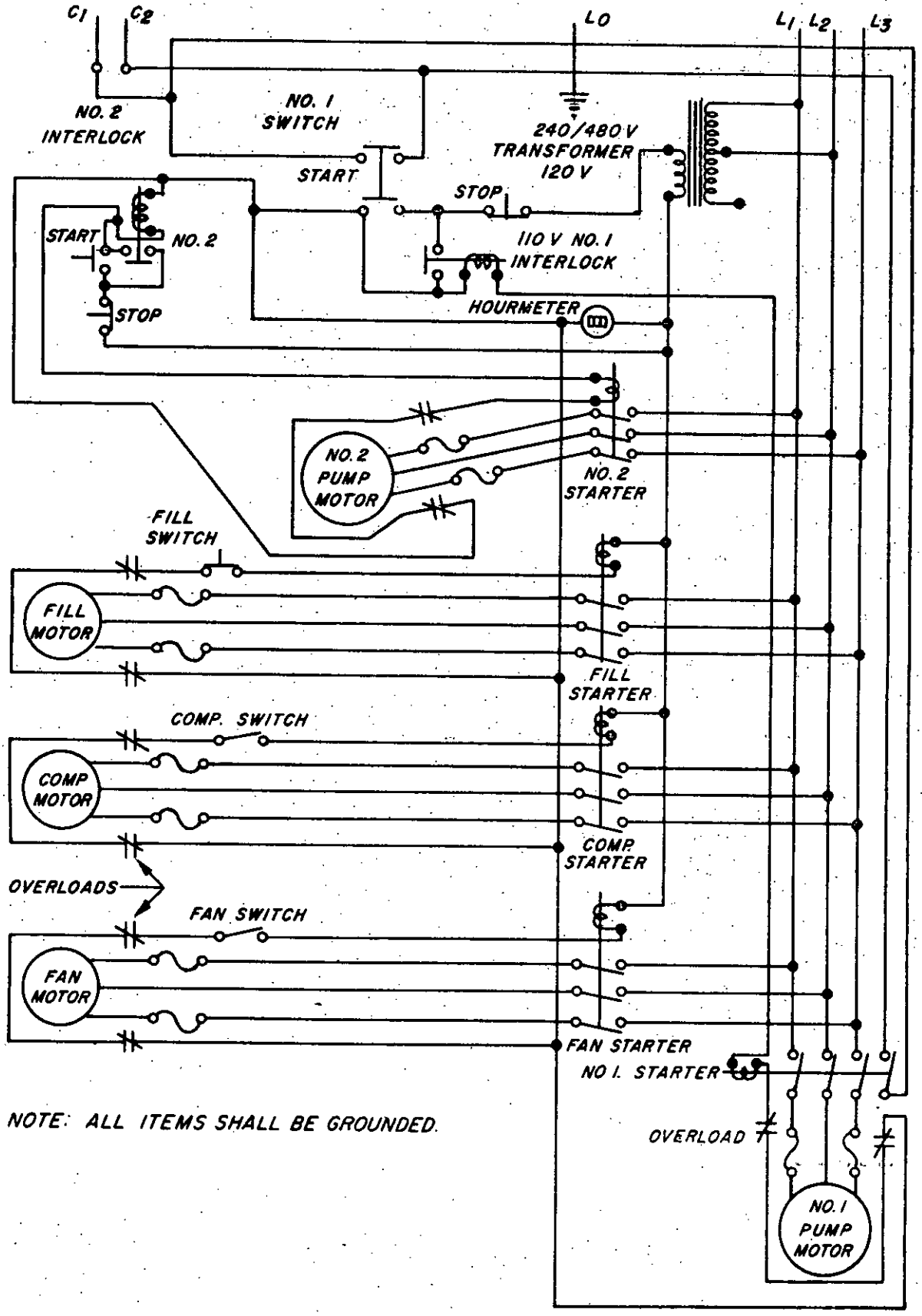
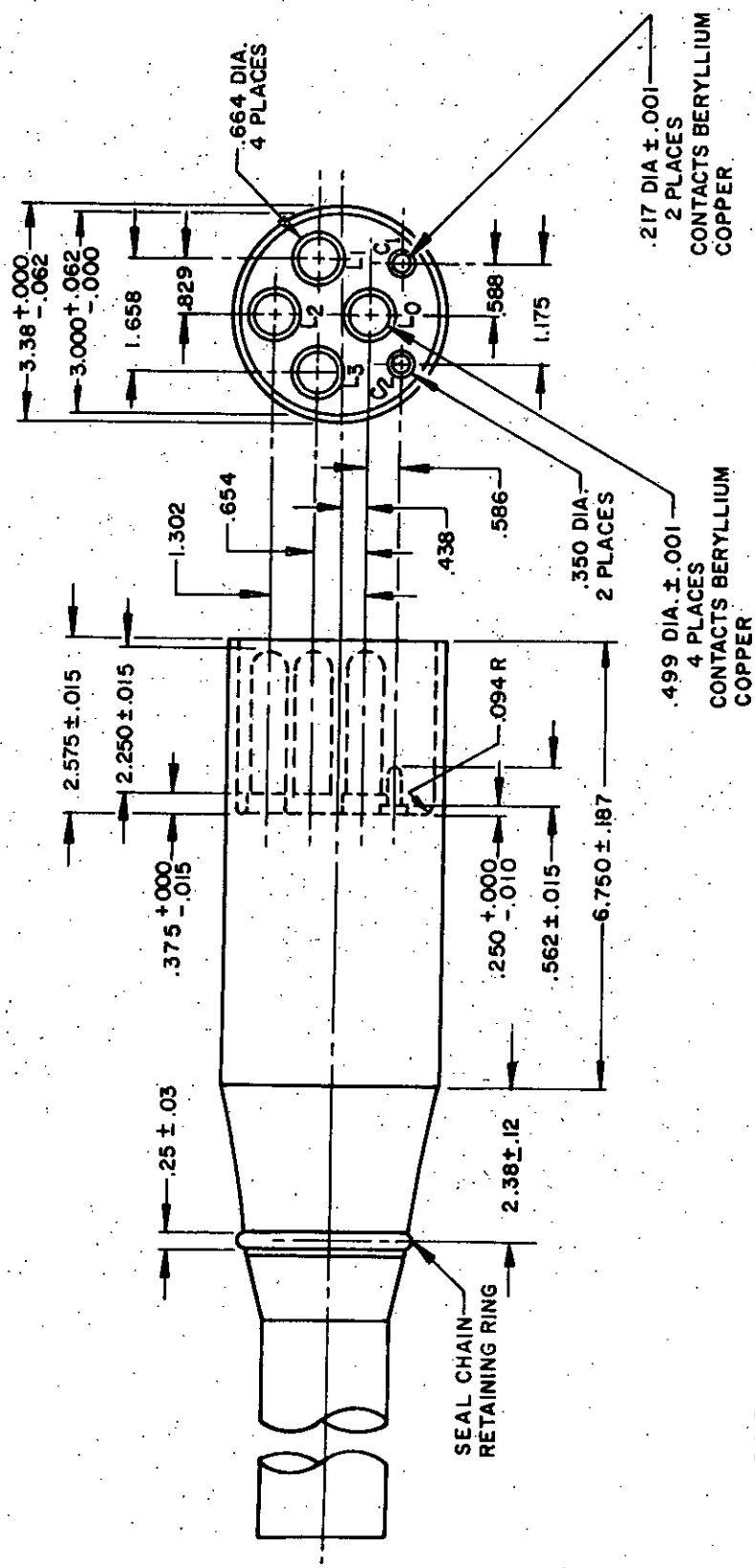


FIGURE 4. ELECTRICAL SYSTEM DIAGRAM

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DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

TOLERANCES: AS SHOWN.

FIGURE 5 CABLE CONNECTOR

*U.S. GOVERNMENT PRINTING OFFICE: 1972-[illegible]/[illegible]

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POSTAGE AND FEES PAID

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS

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SERVICE ENGINEERING DIVISION (MMEOP)
KELLY AFB, TEXAS 78241

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-T_-26017/C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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